Limits – 2

Recurrences solvable in closed form

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Theoretical minimum.

Def. A recurrence is *solvable in closed form* when the sequence can be written through n by elementary functions. Three standard routes: a linear recurrence with constant coefficients; a substitution turning the recurrence into a linear one; and guessing the answer with a proof by induction.

Theorem 1 (linear recurrences). If $a_{n+k} = c_1 a_{n+k-1} + \cdots + c_k a_n$ and the characteristic polynomial $\lambda^k - c_1\lambda^{k-1} - \cdots - c_k$ has roots $\lambda_1, \ldots, \lambda_s$ with multiplicities $m_1, \ldots, m_s$, then $a_n = \sum_i P_i(n)\lambda_i^n$ with $\deg P_i < m_i$. For an inhomogeneous right-hand side of the form $Q(n)\mu^n$ a particular solution is sought in the same shape, with an extra factor n^m if μ is a root of multiplicity m.

Theorem 2 (conjugate surds). If α, β are the roots of $x^2 - px + q$ with integer p, q, then $a_n = \alpha^n + \beta^n$ satisfies $a_{n+1} = pa_n - qa_{n-1}$ and is an integer for all n; the same trick proves integrality of sequences given by a square root, once one guesses the conjugate.

Fact 1. The generating function $F(x) = \sum a_n x^n$ turns a linear recurrence into a rational function; partial fractions return the closed formula, and the pole of smallest modulus gives the asymptotics $a_n \sim C\lambda^n$.

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1. Let c and x_0 be fixed positive numbers and $x_n = \frac{1}{2}\bigl(x_{n-1} + \frac{c}{x_{n-1}}\bigr)$ for $n \geqslant 1$. Prove that the sequence converges and that its limit is $\sqrt{c}$; estimate the speed of convergence and explain why the method is called Newton's.

2. Define (a_n) by $a_1 = 1$ and $a_{n+1} = \frac{1+4a_n+\sqrt{1+24a_n}}{16}$ for $n \geqslant 1$. Find an explicit formula for a_n. *(Hint: substitute $b_n = \sqrt{1+24a_n}$.)*

3. Consider $a_0 = 1$, $a_{n+1} = \frac{3a_n + \sqrt{5a_n^2 - 4}}{2}$ and $b_0 = 0$, $b_{n+1} = a_n - b_n$ for $n \geqslant 0$. Prove that $a_n^2 = b_{2n+1}$ for all n.

4. Find a_n if $a_0 = 1$, $a_1 = 2$ and $a_n = 6a_{n-1} - 8a_{n-2} + n$ for $n \geqslant 2$.

5. (IMC 2025 shortlist) Let $\alpha \in \mathbb{R}$ with $|\alpha| \leqslant 1$, and let $a_0 = 1$, $a_1 = \alpha$, $a_{n+2} = 2\alpha a_{n+1} - a_n$ for $n \geqslant 0$. Prove that $|a_n| \leqslant 1$ for every n. *(Hint: this is exactly the Chebyshev recurrence; put $\alpha = \cos\theta$.)*